

reaches a depth of 80, because the base case of `size` becoming less than 1 is never reached.

Conway's Game of Life Glider

Conway's Game of Life is a famous example of cellular automata. The game's simple rules cause interesting and wildly chaotic patterns to emerge on a 2D grid. One such pattern is a *Glider* consisting of five cells in a 3×3 space:

Python

```
elif DRAW_FRACTAL == 5:
    # Conway's Game of Life Glider:
    third = 1 / 3
    drawFractal(drawFilledSquare, 600,
        [{'sizeChange': third, 'yChange': third},
         {'sizeChange': third, 'xChange': third},
         {'sizeChange': third, 'xChange': third, 'yChange': -third},
         {'sizeChange': third, 'yChange': -third},
         {'sizeChange': third, 'xChange': -third, 'yChange': -third}])
```

The Glider fractal here has additional Gliders drawn inside each of its five cells. The `third` variable helps precisely set the position of the recursive shapes in the 3×3 space.

You can find a Python implementation of Conway's Game of Life in my book *The Big Book of Small Python Projects* (No Starch Press, 2021) and online at <https://inventwithpython.com/bigbookpython/project13.html>. Tragically, John Conway, the mathematician and professor who developed Conway's Game of Life, passed away of complications from COVID-19 in April 2020.

Sierpiński Triangle

We created the *Sierpiński Triangle fractal* in Chapter 9, but our Fractal Art Maker can re-create it as well by using the `drawTriangleOutline()` shape function. After all, a Sierpiński triangle is an equilateral triangle with three smaller equilateral triangles drawn in its interior:

Python